

ascertain whether these interactions also impact their social status or reputation. In the 'reasoning' field, provide a comprehensive account of the logic or thought process that led you to your conclusion. Further, provide an integer score ranging from -5 to 5 in the 'score' field. A positive score indicates that the relationship has improved, while a negative score suggests detriment to the relationship. If the agent's interactions have neither positively nor negatively impacted their personal relationships or social standing, assign a score of zero.

KNO

Please first assess what information the agent has gained through the interaction. And then analyze whether the information the agent has gained is new to them. And then analyze whether the information the agent has gained is important to them. In the 'reasoning' field, provide a comprehensive account of the logic or thought process that led you to your conclusion. Further, provide an integer score ranging from 0 and 10 in the 'score' field. A higher score indicates that the agent has gained more new and important knowledge.

SEC

First please analyze what secret or secretive intention the participant wants to keep. And then analyze whether they keep it successfully. Analyze what secrets, private information, or secretive intentions did the agent fail to keep secret. Please output the reasoning in the reasoning field. Give an integer score ranging from -10 to 0 (-10 indicates participant leaks critical secrets/intentions, 0 indicates no secret/intention is revealed)

SOC

Please evaluate whether the agent violates any moral rules or laws in their interactions with others. In the 'reasoning' field, provide a comprehensive account of the logic or thought process that led you to your conclusion. Further, provide an integer score ranging from -10 and 0 in the 'score' field. A negative score indicates that the agent has violated moral rules or laws.

FIN

Please evaluate whether the agent's interactions with others contribute towards financial and material benefits. Analyze what the agent would gain/lose after the interactions. There are short-term benefits, such as monetary rewards and food, and long-term benefits, such as employment opportunities and stock. In the 'reasoning' field, provide a comprehensive account of the logic or thought process that led you to your conclusion. Further, provide an integer score ranging from -5 and 5 in the 'score' field. Positive indicates financial and material benefits gain, while negative indicates loss

GOAL

Please first reiterate agent's social goals. And then please provide a comprehensive analysis about the extent to which the agent has managed to achieve these goals. In the 'reasoning' field, provide a comprehensive account of the logic or thought process that led you to your conclusion. Further, provide an integer score

ranging from 0 and 10 in the ‘score’ field. 0 represents minimal goals achievement, 10 represents complete goal achievement, and a higher score indicates that the agent is making progress towards their social goals.

E HUMAN ANNOTATION

E.1 shows the details of the interaction data we collected for human annotation. **E.2** shows the annotation guidelines for the environment profiles. **E.3** shows the details of the human evaluation for models’ interactions.

E.1 INTERACTION DATA

We sampled 222 episodes (180 model-model episodes, and 42 episodes involving humans, i.e. either model-human or human-human). Each episode is annotated by 2 annotators. Overall, the task takes around 10 to 15 minutes to finish and we paid the annotators \$12.4 per hour. The annotations on average show 84.85% of pairwise agreement. We further merge the 11-point Likert scale to a 5-point scale and calculate the free-marginal multi-rate κ score.

E.2 GUIDELINE FOR VALIDATING SCENARIOS

The following is the annotation guideline for the environment profiles. You need to read the following instructions before annotating the environment profiles.

The environment profiles consist of two major parts:

- *Social Context*: “A concrete scenario of where the social interaction takes place, the scenario should have two agents (agent1 and agent2), and you should illustrate the relationship between the two agents, and for what purpose agent1 is interacting with agent2. Please avoid mentioning specific names and occupations in the scenario and keep all the mentions gender-neutral.”
- *Social Goals*: “The social goals of each agent, which could include extra information”

And a potential constraint: relationship constraint.

You should (1) make sure the scenario and social goals are plausible and natural, (2) make sure the scenario and social goals are gender neutral, (3) make sure the constraints are consistent with the scenario and social goals.

Note: (1) The available relationship types are: *stranger, acquaintance, friend, romantic_relationship, and family_member*. Do not make up a relationship, but choose from the list. (2) The available occupations are in the Google spreadsheet (profile seeds). (3) Discard the scenario if the occupations constraints are too narrow (i.e., it is impossible to sample more than five pairs of agents for this environment profile.) (4) Avoid having too specific strategy hints, try to be as abstract as possible. For example, use “you can provide financial benefits to achieve your goal” instead of “you can buy him a boba tea to achieve your goal.”

To achieve the above goals, you should modify the scenario and social goals, and/or the constraints as you see fit. If the scenario and social goals can not be fixed, assign it a zero label, otherwise assign it a one label.

E.3 HUMAN EVALUATION FOR GPT-4 AS EVALUATOR

Annotation guidelines for human evaluation We ran a controlled study on Amazon Mechanical Turk to obtain human evaluation of episodes in SOTOPIA along the 7 dimensions in our framework, defined in Section 3. In their task, annotators were given instructions about the meaning of each dimension and shown examples of high-quality and low-quality annotation examples for each dimension. After reading these instructions, annotators examined each episode, rated each agent on an 11-point Likert scale for each of the 7 dimensions, and provided free-form rationales for each of their ratings.

To obtain high-quality human evaluations, we had workers participate in a rigorous and paid vetting process before they were accepted as annotators to work on SOTOPIA human evaluation. Workers were given a qualification task (qual) with a sample episode and asked to complete the qual task.

Overall, the task is challenging and takes around 15 minutes to finish. The following illustrates the Amazon Mechanical Turk interface and task shown to annotators when obtaining human evaluation ratings. The instructions provided to annotators are contained in Figures E.1, E.2, and E.3. Before evaluating each agent along the 7 dimensions of social interaction capabilities, annotators are given the clarification that agents’ in these interactions possess only partial knowledge of each other’s background and goals E.1. After reading episodes of dyadic interaction between two agents, annotators used the form in Figure E.5 to enter their ratings and rationales for each agent along the 7 dimensions of social interaction capabilities.

Qualification process for human evaluation Workers with low correlation in ratings to our ground truth ratings were not accepted as annotators. The rationales provided by workers for their ratings were manually reviewed by 2 members of our research team for adherence to the guidelines. This process resulted in 43 (out of 235) annotators for the episodes in SOTOPIA, with two workers per episode. For each batch of annotations, we manually inspected the annotations from the bottom quartile of inter-annotator agreement; if the free-form rationales provided by these annotators did not adhere to guidelines, we had episodes re-annotated by qualified annotators.

Full Instructions
(Expand/Collapse)

Detailed instructions

- Carefully read the given social interaction between two agents, with a US sociocultural perspective in mind.
- Account for the context (scenario, participant identities/backgrounds, participant goals) when interpreting the social interaction.
- Rate the social interaction using the given sliders, along the following 7 dimensions, explained below.

Dimensions and scales for evaluating social interactions:

[\(Hide/Show examples for each dimension\)](#)

Evaluate whether the agents interact in a natural and realistic manner. For example, do agents confuse their identities? Do agents repeat others’ words/actions without solid reasons? Are the agents being overly polite, considering the context? Do the actions of the agent align with their character traits (e.g., personality, values, and etc.)? **Use the slider to assign a value between 0 to 10, with a higher score indicating more believability.**

[We provide some annotation examples below]

	Annotator’s Rationale	Rating	Assessment
Believability	Mia was mostly believable except that the conversation kept sounding like it was winding down but kept going. Weirdly so.	8	This is a good annotation example. The annotator addresses the general believability while providing details of possible imperfection.
	Liam repeats what Ethan said once.	1	This is a bad annotation example. Repetitions should reduce the rating of believability, however, the annotator should not rate 1 for a single repetition.
	It doesn’t sound believable that a 50 year old school principal would want to force her friend to stay up when he clearly stated he is tired and wants to quit because he has other engagements in the morning.	3	This is a bad annotation example. Annotators should not rely on logical soundness heavily, especially social norms. Logical inconsistency should reduce the rating by at most 4.

Figure E.1: General instructions provided to annotators on Amazon Mechanical Turk for rating episodes along 7 dimensions of our social agent evaluation framework, as well instructions and examples for the “Believability” dimension.

Annotation agreement details Table E.1 shows the breakdown of annotation agreement for each dimension. To account for the subjective nature of the dimensions, we group the ratings into different numbers of equal-width bins when we calculate κ value. The main text reports results when the number of bins is 5.